

Towards density $x^{1-\varepsilon}$ for the $3x + 1$ problem: a program with one open lemma

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Abstract

We assemble, as a single conditional theorem, a program that would push the Krasikov–Lagarias density exponent for the $3x + 1$ problem to its ceiling. *If* one uniform-attenuation lemma holds — the intra-triple variation of the edge certificate is contracted by a factor $\kappa \leq \kappa_{\max} < 1$ at every 3-adic depth and scale — *then* the tempering exponent $\alpha_k \rightarrow 1$, hence $\gamma_k \rightarrow 1$, hence $\#\{n \leq x : \text{orbit reaches } 1\} \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$. The chain has six links, of which five are proved and one is the Open Lemma. Proved links: (1) the tempering law makes each certificate an exactly solvable roulette measure raised to a scalar power; (2) a mass conservation identity, $3W_0 + W_2 + W_8 = 3$ precisely because $2^{\log_2 3} = 3$, forcing subcriticality $a + c \leq 1$; (3) a balance identity, up-flow $= \frac{3}{16}$ = down-flow, so the drift is a pure second-order effect; (4) a directional low-pass, $\min(x + c\mathbf{1}) = \min(x) + c$, making the branch channel a one-way fine-mode filter; (6) an edge-rate theorem, $d\gamma/dq = 1/\ln(4/3) = 3.4761\dots$, which transfers $q \rightarrow 1$ to $\gamma \rightarrow 1$ analytically. The Open Lemma is stated with maximal precision as an LP-duality statement: at the K–L optimum, complementary slackness pins the binding entries, and dual concentration implies a response slope $\leq 1 - \lambda_{\min}$ on intra-triple variation. We give an honest account of what would falsify the program — three “creep” scenarios in which $\kappa \rightarrow 1$ — and why the fine-end saturation (Prop. 23) and depth-flat attenuation (Prop. 35) data disfavor each. We close by contrasting this density-exponent route with Tao’s 2019 almost-all log-density theorem: different quantities, disjoint methods. All statements are labeled proved, verified, measured, or open.

1 Introduction

Let $T(n) = n/2$ for even n , $T(n) = (3n + 1)/2$ for odd n , and let $\pi_1(x) = \#\{n \leq x : \text{the } T\text{-orbit of } n \text{ reaches } 1\}$. The $3x + 1$ conjecture asserts $\pi_1(x) = \lfloor x \rfloor$; Lagarias [2] surveys the problem and Tao [3] proves the strongest known metric result. The strongest known *density* lower bounds come from the difference-inequality method of Krasikov and Lagarias [1]: a finite feasibility certificate at 3-adic depth k yields $\pi_1(x) \geq x^{\gamma_k}$ for all large x . Krasikov and Lagarias obtained $\gamma = 0.84$; the ceiling of the method is $\gamma = 1$, i.e. density $x^{1-\varepsilon}$ for every ε .

Two companion papers develop the ingredients. In [5] we verify certificates to depth $k = 20$ ($\gamma = 0.9146$), identify the certificate eigenvectors as *tempered roulette measures* ($\text{eigvec}_k = \text{roulette}^{\alpha_k}$, $R^2 > 0.99$), prove the edge-rate theorem $d\gamma/dq = 1/\ln(4/3)$, and reduce the full-density statement to a single drift-direction inequality. In [6] we isolate the mechanism of that drift: mass conservation and a balance identity (both forced by $2^{\log_2 3} = 3$), a one-line directional low-pass, and the measured attenuation constant $\kappa_\infty = 0.839 \pm 0.002$, mechanistically identified

as the rigidity of binding constraints in a linear program. The present paper is the *synthesis*: it states the full program as one conditional theorem, marks exactly where each link is proved and where the single remaining lemma is invoked, states that lemma with maximal precision, and confronts it honestly with the ways it could fail.

The result is not a proof of the $3x + 1$ conjecture, nor even of its density form. It is a proof that the density form follows from one clean, local, falsifiable analytic statement — and a demonstration that every other link in the chain, including the two that at first look like the hard analytic content (the transfer from homogenization to density, and the subcriticality of the heterogeneity cascade), is already unconditional.

Labels. Every assertion carries one of four labels. *Proved*: complete mathematical proof (here or in a cited companion). *Verified*: finite exact or floating-point computation, reproducible from stated parameters. *Measured*: statistical fit or extrapolation from computed data. *Open*: neither proved nor finitely checkable. Section 7 collects the ledger.

Notation. $\beta := \log_2 3 = 1.58496\dots$; α is reserved for the tempering exponent. CV denotes coefficient of variation (standard deviation over mean). For $j \geq 1$, $[3^j] := \{m \bmod 3^j : m \equiv 2 \pmod{3}\}$, so $|[3^j]| = 3^{j-1}$. At the edge $(\lambda, \gamma) = (2, 1)$ we write

$$W_0 = \lambda^{-2} = \frac{1}{4}, \quad W_2 = \lambda^{\beta-2} = \frac{3}{4}, \quad W_8 = \lambda^{\beta-1} = \frac{3}{2}. \quad (1)$$

2 The Krasikov–Lagarias edge system

Fix $k \geq 2$ and $\lambda \in (1, 2]$. The nontrivial-tree K–L system $L_k(\lambda)$ [1] seeks $c : [3^k] \rightarrow \mathbb{R}_{>0}$ with, for each $m \in [3^k]$,

$$m \equiv 2 \pmod{9} : \quad c(m) \leq \lambda^{-2}c(4m) + \lambda^{\beta-2} \bar{c}\left(\frac{4m-2}{3}\right), \quad (2)$$

$$m \equiv 5 \pmod{9} : \quad c(m) \leq \lambda^{-2}c(4m), \quad (3)$$

$$m \equiv 8 \pmod{9} : \quad c(m) \leq \lambda^{-2}c(4m) + \lambda^{\beta-1} \bar{c}\left(\frac{2m-1}{3}\right), \quad (4)$$

where $4m$ is reduced mod 3^k , branch targets mod 3^{k-1} , and

$$\bar{c}(t) := \min\{c(t), c(t + 3^{k-1}), c(t + 2 \cdot 3^{k-1})\} \quad (5)$$

is the minimum over the three refinements (the *triple*) of $t \in [3^{k-1}]$. Feasibility of $L_k(\lambda)$ implies $\pi_1(x) \geq x^\gamma$, $\gamma = \log_2 \lambda$, for all large x [1]. We call $m \mapsto 4m$ the *transport* (*up*-) channel and the two branch maps the *down*- channel; all three class maps are bijections of $[3^k]$ resp. onto $[3^{k-1}]$. The right-hand sides define a monotone, positively homogeneous (topical) operator F_λ ; feasibility is equivalent to its nonlinear Perron value being ≥ 1 , and the critical λ_k (the *edge*) is located by bisection, the Perron eigenvector by power iteration.

The verified certificates [5] are

$$(k, \gamma_k, q_k) : \quad (13, 0.8624, 0.9519), (15, 0.8805, 0.9594), (17, 0.8953, 0.9655), \\ (19, 0.9069, 0.9705), (20, 0.9146, 0.9723),$$

where q_k is the min-loss parameter of (13) below. The exponent γ_k climbs steadily but with shrinking increments; the program explains why, and what would have to be shown to push it to 1.

3 The conditional theorem

We first name the missing statement, then state the theorem it closes. The full precision (LP-duality form) is deferred to Section 4.

Open Lemma 1 (Uniform attenuation; informal). *There is a constant $\kappa_{\max} < 1$, independent of depth k and scale P , such that at the Perron fixed point of the critical system $L_k(\lambda_k)$ the minimum (5) contracts intra-triple variation by at most $\kappa_{\max}$: writing $\kappa(P; k)$ for the ratio of the standard deviation of the triple-minimum’s difference-increment to that of its members (Definition 7),*

$$\kappa(P; k) \leq \kappa_{\max} < 1 \quad \text{for all } P \text{ and all } k.$$

Theorem 2 (Conditional full-density bound). *Suppose Open Lemma 1 holds. Then the tempering exponent $\alpha_k \rightarrow 1$, equivalently the min-loss parameter $q_k \rightarrow 1$, equivalently the density exponent $\gamma_k \rightarrow 1$; consequently, for every $\varepsilon > 0$,*

$$\pi_1(x) = \#\{n \leq x : \text{orbit reaches } 1\} \geq x^{1-\varepsilon} \quad \text{for all sufficiently large } x.$$

The rest of Section 3 is the proof of Theorem 2, assembled from six links. For each link we mark **[proved]** or **[Open Lemma]** in boldface at its head, so that the logical dependency is legible at a glance. Links (1)–(4) and (6) are proved unconditionally; the Open Lemma enters only at link (5), where it converts the proved structural low-pass into a quantitative uniform contraction.

3.1 Link (1): tempering law and roulette exact solvability [proved solvability; measured law]

At the level of natural density the Syracuse map is exactly solvable. By the Terras bijection [4], prescribing the first r cascade exponents confines an odd n to one residue class mod $2^{(\text{sum})+1}$, so the w_i are independent geometric(1/2) in density. The stationary measure of the induced residue dynamics mod 3^j — the *roulette measure* $\pi^{(j)}$ — is computable in closed form by downward recursion with no eigenproblem, because the base map $r \mapsto 3r + 1 \bmod 3^j$ depends only on $r \bmod 3^{j-1}$ (Proposition 4 of [5], proved). At depth two,

$$\pi(1, 2, 4, 5, 7, 8) = \frac{(8, 16, 11, 4, 2, 22)}{63} \tag{6}$$

(Theorem 5 of [5], proved). The certificate eigenvectors are pure powers of this measure,

$$\text{eigvec}_k = \text{roulette}^{\alpha_k}, \quad R^2 = 0.9927 \dots 0.9977, \tag{7}$$

with $\alpha_k = 0.8024, 0.8291, 0.8509, 0.8682, 0.8785$ at $k = 13, 15, 17, 19, 20$ (Conjecture T of [5], measured at five depths; the $k = 20$ value was an out-of-sample prediction confirmed to four decimals). The role of this link is structural: it certifies that the entire heterogeneity of the certificate is captured by the scalar deficit $1 - \alpha_k$, so that “homogenization” has a one-parameter meaning. Crucially, the linearization of $L_k(2)$ (replacing the minimum by the triple mean) is exactly the roulette transition operator, whose spectrum is $\{1\} \cup \{|\mu| = 1/4\}$ (verified at $j = 4, 5, 6$; Theorem 21 of [7], Remark 6 of [5]): linear damping toward the roulette is overwhelming, so surviving heterogeneity must be re-injected at each level by the min-nonlinearity. This is the mechanistic frame for links (2)–(5).

3.2 Link (2): mass conservation and subcriticality [proved]

Lemma 3 (Mass conservation at the edge; proved, Lemma 24 of [7]). *With the edge weights (1), the row masses of (2)–(4) are $W_0 + W_2 = 1$ (type 2), $W_0 = \frac{1}{4}$ (type 5), $W_0 + W_8 = \frac{7}{4}$ (type 8), and their average is exactly 1:*

$$3W_0 + W_2 + W_8 = \frac{3}{4} + 3 \cdot 2^{\beta-2} = \frac{3}{4} + \frac{9}{4} = 3, \quad \text{precisely because } 2^\beta = 3. \quad (8)$$

Proof. Direct evaluation; the final step is $3 \cdot 2^{\beta-2} = 3 \cdot 2^\beta / 4 = 9/4$ iff $2^\beta = 3$. $\square$

The difference field of the eigenvector propagates between adjacent trit scales via the exact *offset algebra* [5, 6]: a pure trit offset $\delta = d \cdot 3^P$ satisfies $4\delta = \delta + 3\delta$ (transport: scales $P, P+1$) and $\frac{4\delta}{3} = \frac{\delta}{3} + \delta$ (branch: scales $P-1, P$). Averaging over classes yields the two-coefficient closure (measured, fit error $\leq 1.2\%$ at $k = 13, 15, 17, 19$)

$$\text{CV}_P = a \text{CV}_{P-1} + c \text{CV}_{P+1}, \quad (9)$$

where $\text{CV}_P(k)$ is the mean CV over sibling triples differing only in trit P ($P = 1$ fine end, $P = k - 2$ top).

Corollary 4 (Subcriticality; proved within (9), Cor. of Lemma 24). *Each unit of difference-field mass at scale P is redistributed to $\{P-1, P, P+1\}$ with total weight equal to the row mass of its class (Lemma 3); the minimum (5) is 1-Lipschitz and contributes no expansion. Averaging and applying the triangle inequality bounds the closure coefficients, after absorbing the self-term, by $a + c \leq 1$.*

Measured at $k = 20$: $a + c = 0.9955$. The point is conceptual: the identity $2^\beta = 3$ — the defining equation of the problem — pins the average row mass to exactly 1, so the cascade sits *on* the conservative line and can never grow exponentially. The only remaining freedom is the *direction* of drift along the line.

3.3 Link (3): the balance identity [proved]

Proposition 5 (Balance identity; proved, Prop. 32(i) of [7], Prop. 4 of [6]). *At the edge, the up-flow of offset magnitude (scale $P \rightarrow P+1$, transport) and the down-flow (scale $P \rightarrow P-1$, branch) are exactly equal:*

$$\text{up-flow} = W_0 \cdot \frac{3}{4} = \frac{3}{16} = \bar{w} \cdot \frac{1}{4} = \text{down-flow}, \quad \bar{w} := \frac{W_2 + W_8}{3} = \frac{3}{4}.$$

The zeroth-order scale drift of difference mass is exactly zero.

Proof. Transport $4\delta = \delta + 3\delta$ splits magnitude $4|\delta|$ into $|\delta|$ (scale P) and $3|\delta|$ (scale $P+1$); the up-going fraction is $3/4$ and the transport weight $W_0 = 1/4$, so up-flow $= \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$. Branch $\frac{4\delta}{3} = \frac{\delta}{3} + \delta$ splits magnitude $\frac{4}{3}|\delta|$ into $\frac{1}{3}|\delta|$ (scale $P-1$) and $|\delta|$ (scale P); the down-going fraction is $1/4$. The mean branch weight per class is $\bar{w} = (W_2 + W_8)/3 = \frac{3}{4}$ by mass conservation; so down-flow $= \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}$. $\square$

The consequence is sharp and is the reason link (5) is needed at all: *the drift $c > a$ cannot be a first-moment effect*. Whatever pushes heterogeneity preferentially downward must act through the single structural asymmetry that survives the balance — and there is exactly one.

3.4 Link (4): the directional low-pass [proved core]

Proposition 6 (Directional low-pass; proved, Prop. 34b of [7], Prop. 7 of [6]). *For $x \in \mathbb{R}^3$ and $c \in \mathbb{R}$, $\min(x+c\mathbf{1}) = \min(x)+c$, and $|\min(x)-\min(y)| \leq \max_i |x_i - y_i|$. Hence, decomposing any triple perturbation into its triple-constant (coarse) and intra-triple-varying (fine) components, the minimum transmits coarse modes with slope exactly 1 and is 1-Lipschitz — weakly contractive — on fine modes. Since only the down-channel passes through the minimum, the edge operator transmits all modes freely upward, transmits coarse modes freely, and attenuates precisely the intra-triple-varying modes downward.*

Proof. Every component of x shifts by c , so the least does; and $x_i \leq y_i + \max_j |x_j - y_j|$ for all i , so taking minima and symmetrizing gives the Lipschitz bound. $\square$

This is the entire structural content of the drift: the down-flow of fine modes is starved by a factor $\kappa \leq 1$ (with $\kappa < 1$ whenever the triples are not comonotone in their increments), while the up-flow is free. In (9) this is exactly the statement $c > a$. Proposition 6 is what makes the sign of the drift structural; what it does *not* give by itself is a *uniform* bound $\kappa \leq \kappa_{\max} < 1$ across all depths. Supplying that bound is the Open Lemma (link (5)). The supporting data are strong. Define:

Definition 7 (Attenuation constant). For sibling triples at scale P of the depth- k certificate, let D_i be the member difference-increments under one application of the edge-linearized operator and $D_{\min}$ that of the triple minimum; set $\kappa(P; k) := \text{std}(D_{\min})/\text{std}(D_{\text{member}})$.

Measured (Prop. 35 of [7]; [6]): $\kappa < 0.95$ at every scale and depth, and the top-aligned deep-scale value is *flat* in k :

$$\kappa_{\text{deep}} = 0.841 \text{ (} k=13\text{)}, 0.840 \text{ (15)}, 0.837 \text{ (17)}, 0.838 \text{ (19)} \implies \kappa_{\infty} = 0.839 \pm 0.002. \quad (10)$$

Uniformity in P and k is thus *empirically pinned*; the Open Lemma is the analytic ratification of (10).

3.5 Link (5): uniform attenuation [Open Lemma]

This is the sole conditional step. We invoke Open Lemma 1 (precise form: Open Lemma 10, Section 4) to pass from the proved one-sided low-pass of link (4) to the quantitative, depth-uniform contraction $\kappa \leq \kappa_{\max} < 1$. Combined with links (2)–(4), this fixes the sign *and* a uniform lower bound on the size of the drift:

$$c - a \geq \eta(\kappa_{\max}) > 0 \quad \text{uniformly in } k, \quad (11)$$

where $\eta(\cdot) > 0$ on $[0, 1)$ is the bookkeeping factor converting one-sided low-passing into a gap in the emergent recurrence (the bare-to- effective map; see the caveat in Section 4). On the conservative line $a + c \leq 1$ (Corollary 4) with $c - a \geq \eta > 0$, the decaying root of (9) obeys

$$\theta_k = \frac{a_k}{c_k} \leq \bar{\theta} < 1 \quad \text{uniformly in } k. \quad (12)$$

The measured θ -series is 0.8248, 0.8360, 0.8438, 0.8480, 0.8488 at $k = 13, 15, 17, 19, 20$, converging to $\theta_{\infty} \approx 0.85$ (consistent with $\kappa_{\infty} = 0.839$ to 1.2%; the residual is the up-channel mixing correction). Fine-end saturation (Proposition 11, invoked also in Section 5) supplies the nonzero source that makes (12) nonvacuous.

3.6 Link (6): the edge rate transfers $q \rightarrow 1$ to $\gamma \rightarrow 1$ [proved]

Proposition 8 (Min-loss identity; proved, Thm. 12 of [7], Prop. 9 of [5]). *Let λ_k be critical and c the Perron eigenvector, so (2)–(4) hold with equality. With $S = \sum_{[3^k]} c$, $S_{\min} = \sum_{[3^{k-1}]} \bar{c}$,*

$$1 = \lambda_k^{-2} + \frac{q_k}{3} \left(\lambda_k^{\beta-2} + \lambda_k^{\beta-1} \right), \quad q_k := \frac{3S_{\min}}{S} \in (0, 1], \quad (13)$$

and $q_k = 1$ iff every refinement triple of c is constant, in which case $\lambda = 2$.

The identity follows by summing the edge equalities using that all class maps are bijections; $q_k \leq 1$ with equality iff all triples are flat [7, 5]. Verified to eight decimals at $k = 5, \dots, 11$. It says: *the entire gap between the K–L method and full density is the min-loss $1 - q_k$.*

Theorem 9 (Edge rate; proved, Thm. 19 of [7], Thm. 11 of [5]). *Along (13), with $\gamma = \log_2 \lambda$,*

$$\left. \frac{d\gamma}{dq} \right|_{(\lambda, q)=(2, 1)} = \frac{1}{\ln(4/3)} = 3.4761\dots, \quad \text{so } 1 - \gamma \sim 3.4761(1 - q).$$

Proof. With $F(\lambda, q) = \lambda^{-2} + \frac{q}{3}(\lambda^{\beta-2} + \lambda^{\beta-1}) - 1$, at $(2, 1)$ one computes $F_q = \frac{3}{4}$ and $F_\lambda = \frac{3(\beta-2)}{8}$ (using $2^{\beta-2} = \frac{3}{4}$, $2^{\beta-3} = \frac{3}{8}$), whence $d\lambda/dq = -F_q/F_\lambda = 2/(2 - \beta)$ and $d\gamma/dq = \frac{1}{\lambda \ln 2} \cdot d\lambda/dq = \frac{1}{(2-\beta) \ln 2} = \frac{1}{\ln 4 - \ln 3} = \frac{1}{\ln(4/3)}$. $\square$

The measured ratios $(1 - \gamma_k)/(3.4761(1 - q_k)) = 0.824, 0.847, 0.873, 0.908$ at $k = 13, 15, 17, 19$ increase monotonically to 1 (Corollary of Theorem 9, measured): the transfer from homogenization to density exponent is *analytic*. In particular the empirical constant 0.698 in $\gamma_k = 1 - 0.698(1 - \alpha_k)$ is a finite- k composite, $0.698 = 3.4761 \times (1 - q_k)/\text{CV}_{\text{res}}(k)$, with numerator and denominator each tending to their edge limits.

3.7 Assembly

Chaining the six links:

$$\kappa \leq \kappa_{\max} < 1 \xrightarrow{(2),(3),(4)} \theta_k \leq \bar{\theta} < 1 \implies \text{CV}_{\text{top}}(k) \lesssim \text{CV}_1 \cdot \theta_k^k \rightarrow 0 \xleftrightarrow{(1)} q_k \rightarrow 1 \xrightarrow{(6)} \gamma_k \rightarrow 1. \quad (14)$$

link (5), Open Lemma

The middle equivalence uses the min–mean linearization $E[1 - \min/\text{mean}] = c_1 \text{CV} + c_2 \text{CV}^2$ with $c_1 \in [1.19, 1.45]$ (Prop. 18 of [7], measured), so $1 - q_k = \Theta(\text{CV}_{\text{top}}(k))$, together with $1 - \alpha_k = \text{CV}_{\text{res}}(k)$ (measured). The conclusion $\gamma_k \rightarrow 1$ is exactly $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$, completing the proof of Theorem 2. $\square$

Every link except (5) is unconditional. The Open Lemma is the only place the argument leaves proved ground, and it enters at a single, sharply localized point: the quantitative uniformity of a contraction whose *sign* is already proved (link (4)) and whose *value* is already measured and depth-flat ((10)).

4 The Open Lemma, precisely

We now state the Open Lemma in its maximal-precision form: as a statement about the dual of the Krasikov–Lagarias linear program. The mechanism identification of [6] (measured) is that

the attenuation is not clipping-at-switches but *selection-weighted mean reversion*: the argmin member of each triple systematically co-moves less than the triple average, whether or not the selection changes hands ($P(\text{switch}) \approx 0.66$, flat; the antisymmetric correction ± 0.026 is carried equally by switch and no-switch events, Prop. 34 of [7]). The LP reading is exact: the min-selected member is the *binding* constraint of the row that reads it, and binding constraints are pinned by complementary slackness while slack members float.

Open Lemma 10 (Uniform attenuation at the fixed point; LP-duality form). *Let c_k be the Perron eigenvector of the critical system $L_k(\lambda_k)$, regarded as the optimal solution of the K - L linear program at $\lambda = \lambda_k$, and let $y_k \geq 0$ be an optimal dual solution. For $t \in [3^{k-1}]$ let $b(t) \in \{1, 2, 3\}$ be the binding (argmin) index of the triple over t in (5), and let $\lambda_{\min}(t)$ be the dual pinning strength of the binding entry, i.e. the sensitivity coefficient*

$$\lambda_{\min}(t) := 1 - \frac{\partial D_{\min}(t)}{\partial \bar{D}(t)},$$

the shortfall between unit response of a free member and the response of the binding member to a coarse (triple-constant) perturbation $\bar{D}(t)$ of the eigenvector. Then there exists $\lambda_\star > 0$, independent of t , P , and k , such that at the fixed point

$$\lambda_{\min}(t) \geq \lambda_\star \quad \text{whenever the triple over } t \text{ is non-degenerate,}$$

provided the intra-triple coefficient of variation is bounded below,

$$\inf_k \text{CV}_1(k) > 0. \tag{15}$$

Equivalently, the response slope of the down-channel on intra-triple variation satisfies $\kappa(P; k) \leq 1 - \lambda_\star =: \kappa_{\max} < 1$ uniformly.

Why complementary slackness pins the binding entries. At the critical λ_k the certificate saturates (2)–(4); the down-channel of a row of type 2 or 8 reads its target triple only through the minimum (5), so the KKT stationarity of the program attaches a strictly positive dual multiplier to the argmin entry of every such triple and zero to the two slack entries (complementary slackness). Coarse perturbations of the eigenvector — the triple-constant modes of link (4) — move slack and binding entries alike with slope 1 through the *level* term $\lambda^{-2}c(4m)$ (transport), but the binding entry additionally feeds the saturated down-constraint, whose multiplier resists its motion. The resulting response deficit is exactly $\lambda_{\min}(t)$; dual concentration — the claim that the multiplier mass does not thin to zero on any binding entry as $k \rightarrow \infty$ — is what makes $\lambda_{\min}(t) \geq \lambda_\star > 0$ uniform.

Why dual concentration implies slope $\leq 1 - \lambda_{\min}$. Sensitivity analysis of the LP: perturb the right-hand data by a coarse field $\bar{D}$. Slack members respond with slope 1 (Proposition 6, coarse channel). A binding member's response is reduced by its pinning strength, so its slope is $1 - \lambda_{\min}(t)$. Averaging the member responses gives the triple-minimum response slope $\partial D_{\min}/\partial \bar{D} = 1 - \lambda_{\min}(t)$, hence $\kappa(P; k) \leq \max_t(1 - \lambda_{\min}(t)) \leq 1 - \lambda_\star$. Non-degenerate triple gaps — guaranteed by (15) — keep the binding set stable under small perturbations, so the reduction is uniform rather than washing out in the limit.

Where the specific arithmetic must enter. Steps “slack $\Rightarrow$ slope 1” and “binding $\Rightarrow$ slope $1 - \lambda_{\min}$ ” are standard LP sensitivity facts once dual concentration is available. The unproved content is the *uniform lower bound* $\lambda_\star > 0$ on the dual pinning strength. This is where the identities of links (2)–(3) must be used: mass conservation (Lemma 3) normalizes the total dual mass to a fixed budget, and the balance identity (Proposition 5) prevents that budget from draining to either end of the scale hierarchy — neither the fine end (where it would be killed by the vanishing of CV_P , but (15) forbids the source vanishing) nor the top (where it would leave interior triples slack). A proof of Open Lemma 10 is therefore a proof that the defining arithmetic $2^\beta = 3$, which pins the mass and balance identities, also pins the dual mass away from zero on binding entries.

The bookkeeping caveat. The recurrence (9) is *emergent*, not a per-application transfer law: a single application of the edge operator is scale-scattering (60–68% of any mode’s output energy lands on the finest scale, regardless of input scale; measured, [6]), so (9) describes the Perron fixed point, not one step. Consequently the conversion $\kappa \leq \kappa_{\max} \Rightarrow c - a \geq \eta > 0$ (the factor η in (11)) is itself a fixed-point statement, and the “bare” coefficients read off naively from the one-step weights *invert* the drift (measured negative result, [6]). This is precisely why the Open Lemma is posed at the fixed point and its proof route is variational (LP-duality) rather than iterative. We flag this honestly: Open Lemma 10 closes the chain modulo the bare-to-effective bookkeeping, and a fully rigorous write-up must discharge both together.

Measured targets the proof must reproduce. $\kappa_\infty = 0.839 \pm 0.002$; $\lambda_{\text{clip}} := -\text{cov}(\bar{D}, R)/\text{Var}(\bar{D})$ growing $0.085 \rightarrow 0.140$ with depth; the antisymmetric correction ± 0.026 carried equally by switch and no-switch events; and the exact clipping decomposition $\kappa^2 = \text{Var}(\bar{D})/V + 2\text{cov}/V + \text{Var}(R)/V$ with the entire attenuation in the covariance term ($2\text{cov}/V = -0.170 \rightarrow -0.257$) and $\text{Var}(R)/V \leq 0.028$ (verified exact identity, [6]).

5 What would falsify this

The program is falsifiable, and its weakest joint is precisely the Open Lemma: everything reduces to $\kappa \leq \kappa_{\max} < 1$ uniformly. There is exactly one way the conditional theorem can be vacuous: the attenuation *creeps* to 1 as depth grows, $\kappa(P; k) \rightarrow 1$, so that $\theta_k \rightarrow 1$ and the γ_k -increments, though positive, sum to a limit strictly below 1. We enumerate the concrete creep scenarios and confront each with data.

Creep A: triples comonotonize. If the three members of each refinement triple became asymptotically comonotone in their increments, the minimum would track the mean and $\kappa \rightarrow 1$. This is the direct negation of link (4)’s nontriviality. It is disfavored by *fine-end saturation*:

Proposition 11 (Fine-end saturation; measured, nine depths, Prop. 23 of [7]). *The finest-level triple CV of the certificate satisfies*

$$\text{CV}_1(k) = 0.5136 - 0.337(0.910)^k$$

with residual 10^{-6} across $8 \leq k \leq 20$: geometric convergence to the finite, strictly positive limit $\text{CV}_1^\infty \approx 0.514$.

Comonotonization requires $CV_1 \rightarrow 0$; the measured saturation at a nonzero limit forbids it. The raw intra-triple variation on which the low-pass acts is permanently supplied, and its limit (0.514) is not marginal but order unity. For Creep A to occur, the nine-depth fit — residual 10^{-6} , a clean geometric law with a positive asymptote — would have to be an artifact that reverses beyond $k = 20$, turning the asymptote downward. No mechanism for such a reversal is known, and the linear edge spectrum (damping $1/4$, injection at each level; link (1)) predicts a floor, not a decay.

Creep B: attenuation drifts up with depth. Even with CV_1 saturated, κ could in principle drift toward 1 if the selection-weighted mean reversion weakened at large k . This is disfavored by the *depth-flatness* of the deep-scale attenuation:

Proposition 12 (Kappa uniformity; measured, four certificates, Prop. 35 of [7]). $\kappa(P; k) < 0.95$ at every scale and depth; the top-aligned deep-scale value is flat in k : $\kappa_{\text{deep}} = 0.841/0.840/0.837/0.838$ at $k = 13/15/17/19$, giving $\kappa_{\infty} = 0.839 \pm 0.002$.

This is the empirically decisive datum. Every *other* measured quantity of the program drifts with depth — α_k , q_k , θ_k all move monotonically — but the attenuation does not: it is a *constant* of the cascade, not a running coupling. If anything the deep-scale value drifts very slightly *downward* ($0.841 \rightarrow 0.838$), the wrong direction for Creep B. For Creep B, κ_{deep} would need to turn around and climb through 0.95 and onward to 1; four certificates spanning $3^{12} \rightarrow 3^{18}$ classes show no such trend.

Creep C: the bare-to-effective map degenerates. The subtlest scenario: κ stays bounded below 1, yet the emergent gap $c - a$ (factor η in (11)) shrinks to 0 because the fixed-point bookkeeping converting one-sided low-passing into a profile drift degenerates. This is not excluded by κ -data alone — it is the honest soft spot flagged in Section 4. It is disfavored, but only indirectly, by the *measured θ -series* itself: $\theta_k = 0.8248 \rightarrow 0.8488$ with increments shrinking by ≈ 0.6 per step, a textbook geometric approach to a limit ≈ 0.85 , comfortably below 1 and consistent with $\kappa_{\infty} = 0.839$ to 1.2%. A degenerating bare-to-effective map would show $\theta_k \rightarrow 1$; the observed convergence is to 0.85. Still, we rate Creep C the residual risk: the balance identity (link (3)) guarantees the drift is second-order, and a second-order effect could in principle vanish faster than measured without contradicting any proved statement. Closing Creep C is exactly the content of proving the bookkeeping alongside the Open Lemma.

Summary of falsifiability. The framework makes a sharp $k = 21$ prediction [5, 6]: $\gamma \approx 0.918$, $\theta \approx 0.850$, $\kappa_{\text{deep}} = 0.839 \pm 0.002$, fine-end saturation rate 0.910, and a tempering fit $R^2 > 0.9977$. Any material deviation — κ_{deep} rising, CV_1 turning down, θ overshooting 0.86 — falsifies the program at its stated form. The data at five depths disfavor Creeps A and B decisively and Creep C indirectly; none is excluded with the force of a proof, which is exactly what the Open Lemma would supply.

6 Relation to Tao (2019)

Tao [3] proves that almost all Collatz orbits attain almost bounded values: for any function $f(x) \rightarrow \infty$, the set of n whose orbit ever drops below $f(n)$ has *logarithmic* density 1. This is the

strongest known metric result on the problem. It is worth being precise about how it relates to the present program, because the two are easy to conflate and are in fact orthogonal.

Different quantity. Tao controls a *logarithmic-density* statement about *almost all* starting values — an almost-everywhere / measure result with an exceptional set of log-density zero (of unspecified, possibly large, natural density). The present program targets a *natural-density exponent*: an explicit lower bound $\pi_1(x) \geq x^\gamma$ on the *count* of convergent integers, pushing γ from 0.84 toward 1. Neither statement implies the other. Tao’s “almost all in log-density” does not yield any power-law count x^γ with γ near 1 in natural density; conversely $\gamma \rightarrow 1$ would give $\pi_1(x) \geq x^{1-\varepsilon}$ but says nothing about the remaining $\leq x^{1-\varepsilon}$ integers, and in particular does not reach the almost-all conclusion in Tao’s sense. Both fall short of the conjecture $\pi_1(x) = \lfloor x \rfloor$, from opposite directions.

Different method. Tao’s proof is *analytic and probabilistic*: he studies the Syracuse random variable on 3-adic-like cyclic groups, establishes quantitative equidistribution (a Fourier / stability estimate for the Syracuse valuation distribution), and runs a first-moment / weighted argument over a random starting value. The present program is *spectral and structural on a finite linear feasibility system*: it verifies exact Krasikov–Lagarias certificates, identifies their eigenvectors as tempered roulette measures, and reduces the exponent limit to an LP-duality statement about a nonlinear Perron fixed point. The shared object — the geometric- w / Syracuse valuation law and its residue dynamics mod 3^j (our roulette measure, link (1); Tao’s Syracuse distribution) — is the same underlying arithmetic, but it is used at opposite ends: Tao averages it against a random orbit to get a metric almost-all statement; we temper it to fit a deterministic certificate and read off a density exponent.

Complementarity. The two are complementary rather than competitive. Tao’s result says the *typical* orbit descends; ours (conditionally) would say a *near-full-density set* provably reaches 1 with an explicit exponent. A resolution of the full conjecture presumably needs the pointwise content that neither reaches — for us, that the countable integer thread cannot ride the measure-zero survivor set (the “ultra-compressible needle” of [7], Prop. 13); for the metric route, the upgrade from log-density to natural density and from almost-all to all. The programs illuminate different faces of the same wall.

7 Honest status

Proved	Link (1) roulette solvability + closed form mod 9 (6) ([5]); link (2) mass conservation $3W_0 + W_2 + W_8 = 3$ and subcriticality $a + c \leq 1$ (Lem. 3, Cor. 4); link (3) balance identity, up-flow = $\frac{3}{16}$ = down-flow (Prop. 5); link (4) directional low-pass $\min(x + c\mathbf{1}) = \min(x) + c$, 1-Lipschitz on fine modes, down-channel only (Prop. 6); link (6) min-loss identity (Prop. 8) and edge rate $d\gamma/dq = 1/\ln(4/3)$ (Thm. 9); feasibility $\Rightarrow$ density bound ([1]).
Verified	Certificates $k = 13\text{--}20$, $\gamma = 0.8624\text{--}0.9146$; identity (13) to 8 decimals at $k = 5\text{--}11$; linear edge spectrum $\{1\} \cup \{ \mu = 1/4\}$ at $j = 4, 5, 6$; pointwise transport identity (residual 2.8×10^{-4}); clipping decomposition as an exact identity; Gaussian Monte Carlo mismatch; Terras cylinder densities.
Measured	Tempering law $\alpha_k \uparrow$, $R^2 > 0.99$ (link (1)); $1 - \alpha_k = \text{CV}_{\text{res}}$; constant 0.698; edge-rate ratios $0.824 \rightarrow 0.908$; closure (9), $a + c = 0.9955$ at $k = 20$; θ -series $0.8248 \rightarrow 0.8488$; $\kappa < 0.95$ everywhere, $\kappa_\infty = 0.839 \pm 0.002$, depth-flat (Prop. 12); $\lambda_{\text{clip}} = 0.085\text{--}0.140$, antisymmetric ± 0.026 , not switch-exclusive; fine-end saturation $\text{CV}_1 \rightarrow 0.514 > 0$ (Prop. 11); scale-scattering 60–68% to digit 0; one-step reconstruction inversion; out-of-sample hits at $k = 19, 20$.
Open	Open Lemma 10: uniform attenuation $\kappa \leq \kappa_{\text{max}} < 1$ at the fixed point (LP-duality: dual concentration on binding entries $\Rightarrow$ response slope $\leq 1 - \lambda_{\text{min}}$), given $\inf_k \text{CV}_1 > 0$; plus the bare-to-effective bookkeeping converting one-sided low-passing into $c - a > 0$.

The shape of the situation is unusual and worth stating plainly. The best known rigorous handle on the density of $3x + 1$ convergence — the Krasikov–Lagarias certificate — is, to $R^2 > 0.99$ at a billion classes, one exactly solvable measure raised to one scalar power (link (1)). The distance of that power from 1 *is* the distance of the density exponent from 1, through an explicit derivative $1/\ln(4/3)$ (link (6)). The defining constant of the problem, $2^\beta = 3$, pins the certificate’s heterogeneity cascade to the conservative line (link (2)) and balances its zeroth-order scale flow exactly (link (3)); the only asymmetry left is that the downward channel passes through a minimum, which transmits coarse structure perfectly and attenuates fine structure (link (4)); and that attenuation is measured to be a depth-independent constant, $\kappa_\infty = 0.839 \pm 0.002$, mechanistically identified as the rigidity of binding constraints in a linear program. Nothing in the chain is mysterious. One variational inequality — the Open Lemma, LP-duality form — remains to be written down with full rigor, and the negative results show it must be proved at the fixed point, not by iteration.

Reproducibility

All certificates, verification logs, and the scripts generating every number in this paper are archived in the research dossier accompanying this work (directories `certificates/`, `scripts/`, `findings/`), and in the two companion papers [5, 6].

Acknowledgment of method

The computations, fits, and drafts were produced in an extended human–AI collaboration; every proved statement above has a self-contained proof in the text or in a cited companion, independent of that provenance.

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